



IGNITION CONTROLLED POWER MANAGEMENT

Explainer

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PURPOSE

Explain the Ignition controlled power management implemented in Nomad Digital CCU platforms

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QUALITY MANAGEMENT SYSTEM STANDARDS



The Quality Management System of Nomad Digital Ltd has been certified to the ISO 9001:2015 standard. The standard is applicable to the management of activities associated with the design, development, deployment, projects and operational support of mobile wireless data solutions for the transportation sector. These solutions include on-vehicle systems, fixed network systems, backhaul systems, network operating centre services, hardware and software.

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Ignition controlled power management

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This document contains explanation of the Ignition controlled power management implemented in all recent Nomad Digital designed CCU platforms (R4600-2Ax, R4600-3Ax, R5001C).

1.1 What is Ignition controlled Power Management?

Ignition controlled Power Management is a feature to power up and down the CCU platform using a signal from the train. The signal is called an Ignition signal, which is generated from the Train Computer when the locomotive driver inserts the key in the driver's desk in the driver's cabin and turns the Train unit "ON" [Same as switching on Ignition in a car and the light, audio system etc. is turned on].

When the locomotive driver turns off the train and removes the key, a timer is started which keeps power on the train unit for a duration enough to walk to the opposite end of the train unit and put the key in the driver's desk there. This is often used when turning a train unit at a terminal station. During that period the Ignition signal remains high.

If the locomotive driver leaves the train after removing the key from the driver's desk, the Ignition signal drops when the timer has expired. When various systems have the Ignition signal removed, they must power down to avoid draining the train batteries / consuming unnecessary power. This is especially critical for Diesel powered trains.

Our CCU/ODC systems with Ignition control will power up when the Ignition signal enters "high" level, and when the Ignition signal enters "low" level the system will initiate system shutdown either instantly or after current tasks has been completed. The maximum operational time after the Ignition signal has dropped is an important parameter for our systems.

1.2 Why do we need Ignition Controlled Power Management in our CCUs?

All systems can crash due to a sudden power outage if the power is removed at a point in time where the HW/SW system is vulnerable e.g. when SW removes write protection to store important data, during FW updates of host OS or Modem modules, or if the power outage pull out energy from the system, the HW can brown out and corrupt low-level FW data stored in flash storage devices etc.

By adding Ignition controlled power management Nomad Digital can guarantee the stability of the solution which is strongly recommended for following reasons:

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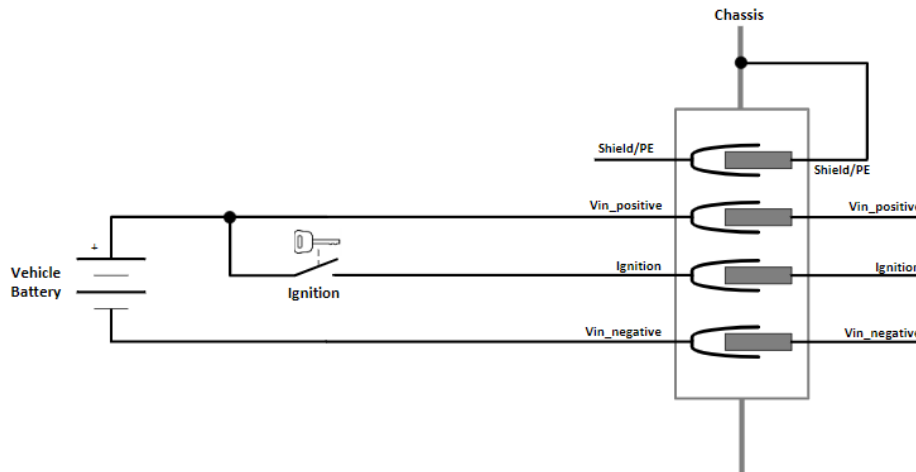
- Avoid file system errors on storage devices. Nomad Connect itself has already been designed to limit writing and reduce this risk. However, safe shutdown is required if applications are hosted, which frequently write to storage. For instance, CCTV, On Board Datacentre with virtualization or logging applications.
- When our CCU platform is used as ODC hosting mission critical systems as e.g. Passenger Information system, we must be connected to a higher priority power circuit which usual require Ignition power management.
- Recent high-performance cellular modems are based on Software Defined Radios (SDR) and can contain up to three CPU systems running on the same program storage and RAM, so they must be shutdown according to a special process, which can only be achieved when using Ignition Power Control. Ignition control is therefore required for configurations with recent 4G LTE Advanced modems respectively 5G NR modems.
- The SIM modules contain a small computer system to handle encryption and storage of network parameters. Due to its open file system, allowing the carrier network to write control parameters etc. to the SIM module, it need to be shut down gracefully through the connected Cellular modem using the Ignition power control to mitigate corruption. As the SIM module is a security device, any corruption in its file system will activate the tampering protection fuse which will self-destroy the SIM. Hence it is very important to ensure that the SIMs are gracefully shutdown, this is also an ETSI requirement to connect to European Mobile Carrier networks – there are similar mandatory requirements for the US Carrier certification of products.
- If our system is connected to permanent power, a longer shutdown can be configured. This offers additional advantages:
 - Allow to keep the Network/Wi-Fi service in full operation while changing the train consist during a journey, or when changing the traction power system etc.
 - Allows for bidirectional synchronization of data between Train and shore before the communications system is closing.
 - Allows for completing SW upgrade before system is closing.

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1.3 How is Ignition signal connected to our CCU

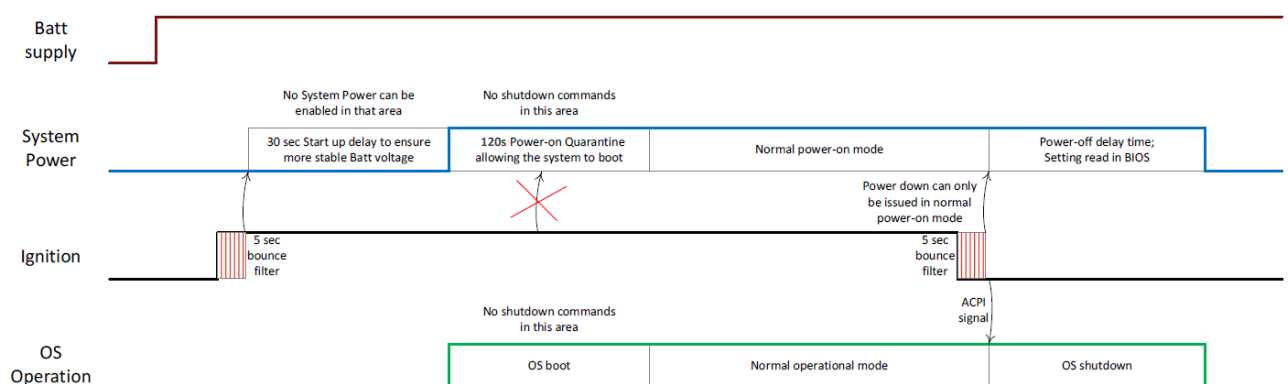
The power supply and ignition signal should be connected as per below. Both the battery and Ignition input are fully isolated and floating. Ignition signal must be referenced to the Train Battery supply.



The Ignition switch is typical an output relay in the train computer, but in old rolling stock the Ignition signal is coming from the Ignition key switch.

1.4 Ignition Management design

An Ignition Management system must be designed to cope with various abnormal situations to remain stable in all conditions of the train supply/Ignition signal. It is not uncommon to see standard products with ignition controlled power on/off driving around on a vehicle switched off due to contact bounce on the Ignition signal. Our Ignition Management system is designed according to the below basic rules:



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1.5 Ignition control management in our CCUs

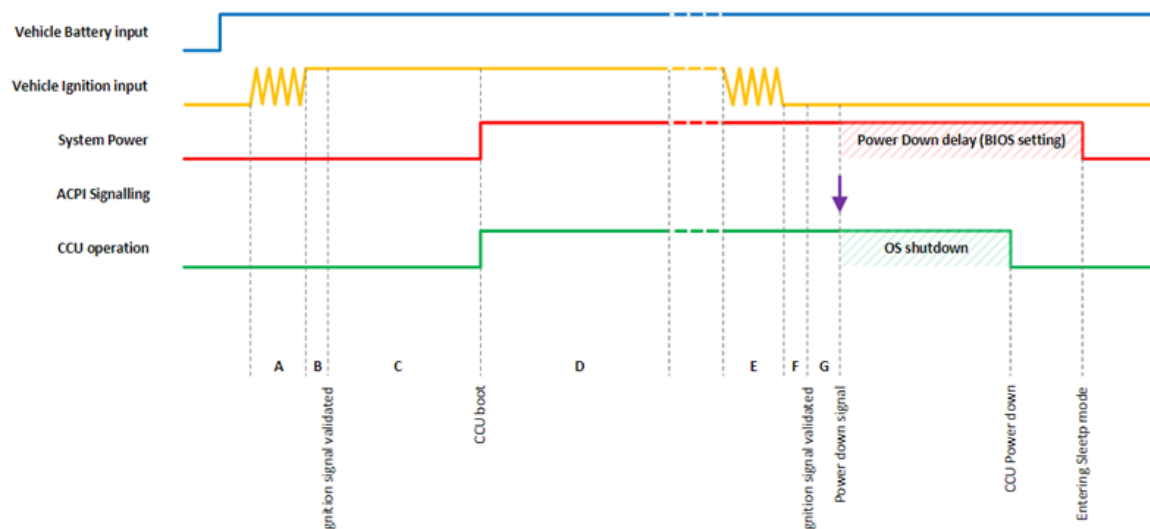


Figure: Power-Up and Power-Down Cycle

- **A:** Contact bounce filtering; waits until ignition signal stabilizes
- **B:** 5 sec Signal monitoring; validates that the signal level remains stable for 5 seconds
- **C:** 30 sec Power on delay; delay CCU boot 30 seconds
- **D:** 120 sec Inhibit Ignition shutdown; ensures no ACPI shutdown signals send before System SW is completely booted
- **E/F:** Contact bounce filtering, waits until ignition signal stabilizes
- **G:** 5 sec Signal monitoring; validates that the signal level remains stable for 5 seconds before ACPI signalling to CCU SW
- **OS shutdown:** time depends on the application, which can be a duration from some seconds to a few minutes.
- **The Power Down delay** is configurable [1, 4, 10, 15, 30 minutes]. Please note that the CCU System cannot be restarted before the Power Down delay has expired.

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1.6 How is the Ignition controlled power on/off working in details?

Startup:

- When the Ignition signal is applied, the CCU System first validates stability of the Ignition signal to ensure there is no relay contact bouncing on the Ignition signal input (very fast changing high, low, high....). The CCU System will not continue before the Ignition signal state is stable at high level and fulfil the calculated threshold level based on the applied Battery voltage.
- The ignition threshold voltage is referenced to the applied battery voltage (Vin_positive), being 70% of Vin_positive. For instance, a battery voltage of 110VDC will define the threshold value on the ignition pin to $0.7 \times 110 = 77\text{VDC}$. Same procedure for the other standard battery voltages 24VDC, 36VDC, 72VDC according EN50155.
- When Ignition signal input is validated “high”, then a 5sec monitoring period is started which allows the Ignition signal to be cancelled without start of the CCU System. Beyond that point the CCU System will boot even if the Ignition signal drops again.
- If the Ignition signal remains high after the 5sec monitoring period, a 30sec startup delay is activated. The startup delay ensures that the Vehicle Power supply has stabilised before the CCU is starting.
- When the 30 sec startup delay is expired, power is applied to the hosted system components and the boot process will start.
- When power is applied to the hosted system components, a 120sec quarantine timer is started. Within the 120sec the CCU System will ignore changes to the ignition input, means no shutdown request possible. Reason is that the CPU system(s) cannot receive the ACPI shutdown signalling before the system is fully booted.

Shutdown:

- When the Ignition signal is removed, or voltage level is $< 20\%$ of Vin_positive (e.g. $0.2 \times 110 = 22\text{V}$) it will pass the same validation process as during startup with relay contact bounce filtering and 5sec monitoring period to allow for a cancel of the Ignition shutdown request.
- When Ignition shutdown is validated, the CCU System will send a ACPI shutdown command to CPU System(s) and continue applying power to the system according to the configured power off delay settings (User defined). During the power off delay, the CPU System(s) will complete their tasks and do a graceful shutdown.

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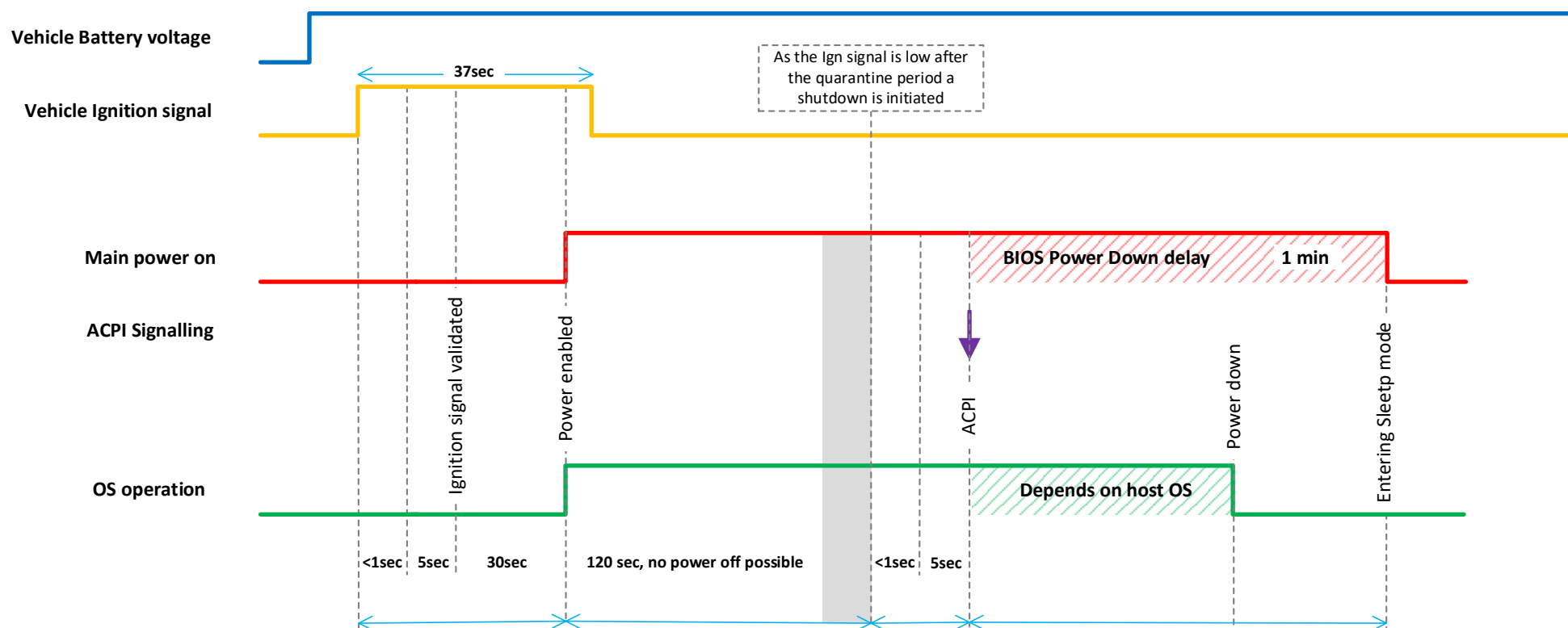
- After the graceful shutdown is completed, the CPU system(s) are powered down, and when the power off delay timers is expired the power to the System slots will be removed. Then the CCU Rack chassis enters sleep-mode (very low power consumption), only monitoring the state on the Ignition signal.
 - If connected to permanent power, the system will restart when ignition signal returns.
 - Recommended to wait 2 minutes - after Ignition signal drops - before disconnecting the train battery supply. The minimum for Nomad Connect is 1 minute.

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1.7 What happens in different situations?

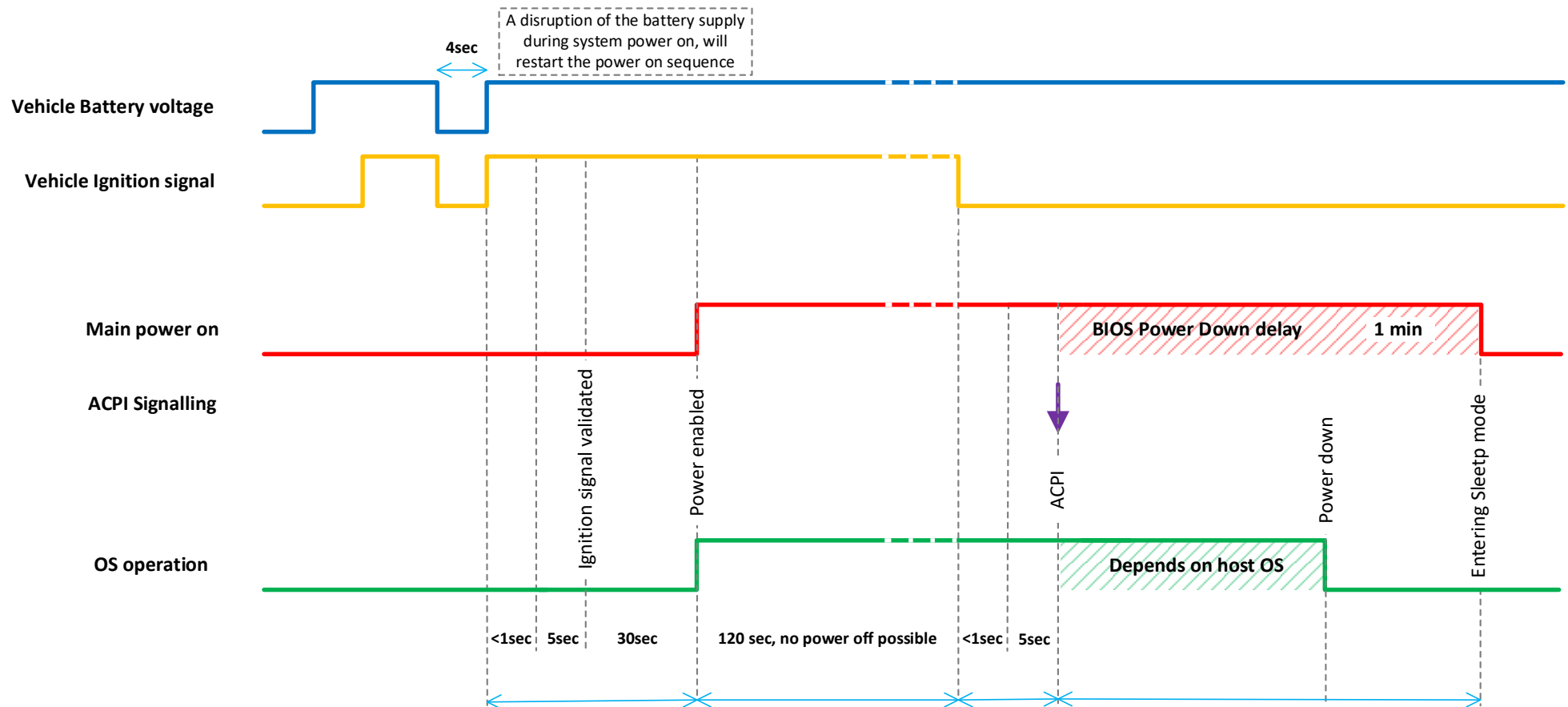
If Ignition signal drops after the signal validation period of approx. 35sec., the CCU System will boot and after the 120sec quarantine period, it will power down after the configured power down delay has expired.



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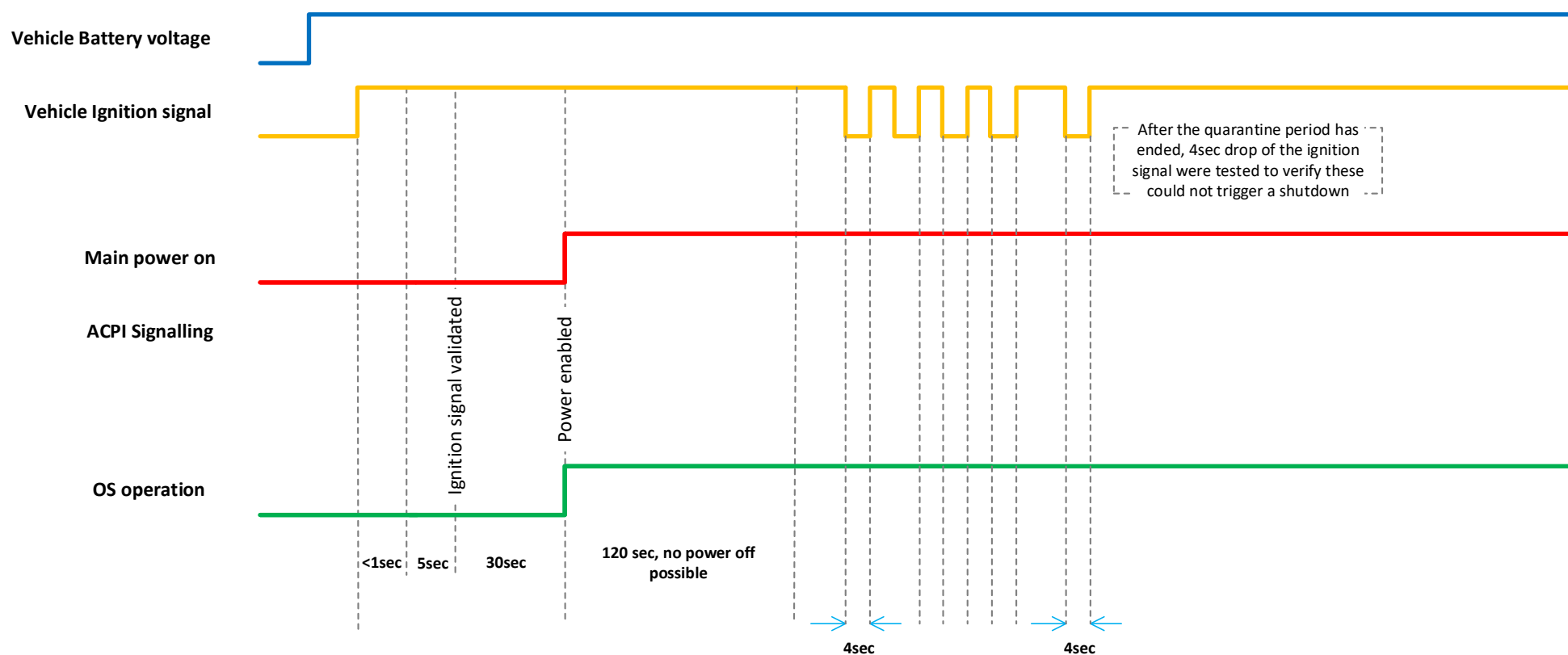
If the Ignition signal from the train drops out for $\leq 4\text{sec}$ during the signal evaluation period, the power on sequence will be restarted.



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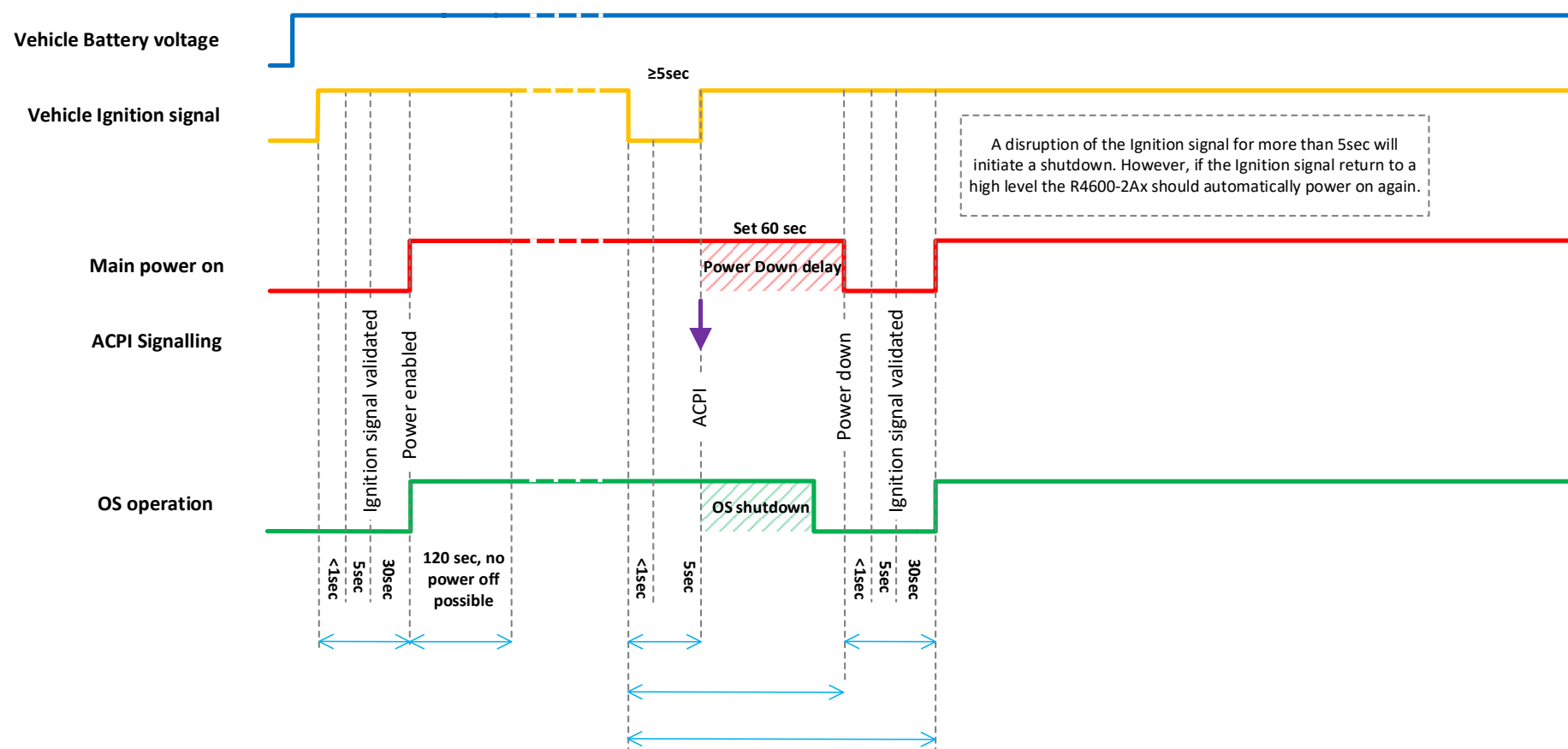
If the Ignition signal flickers with a low period ≤ 4 sec no CCU System shutdown will be initiated.



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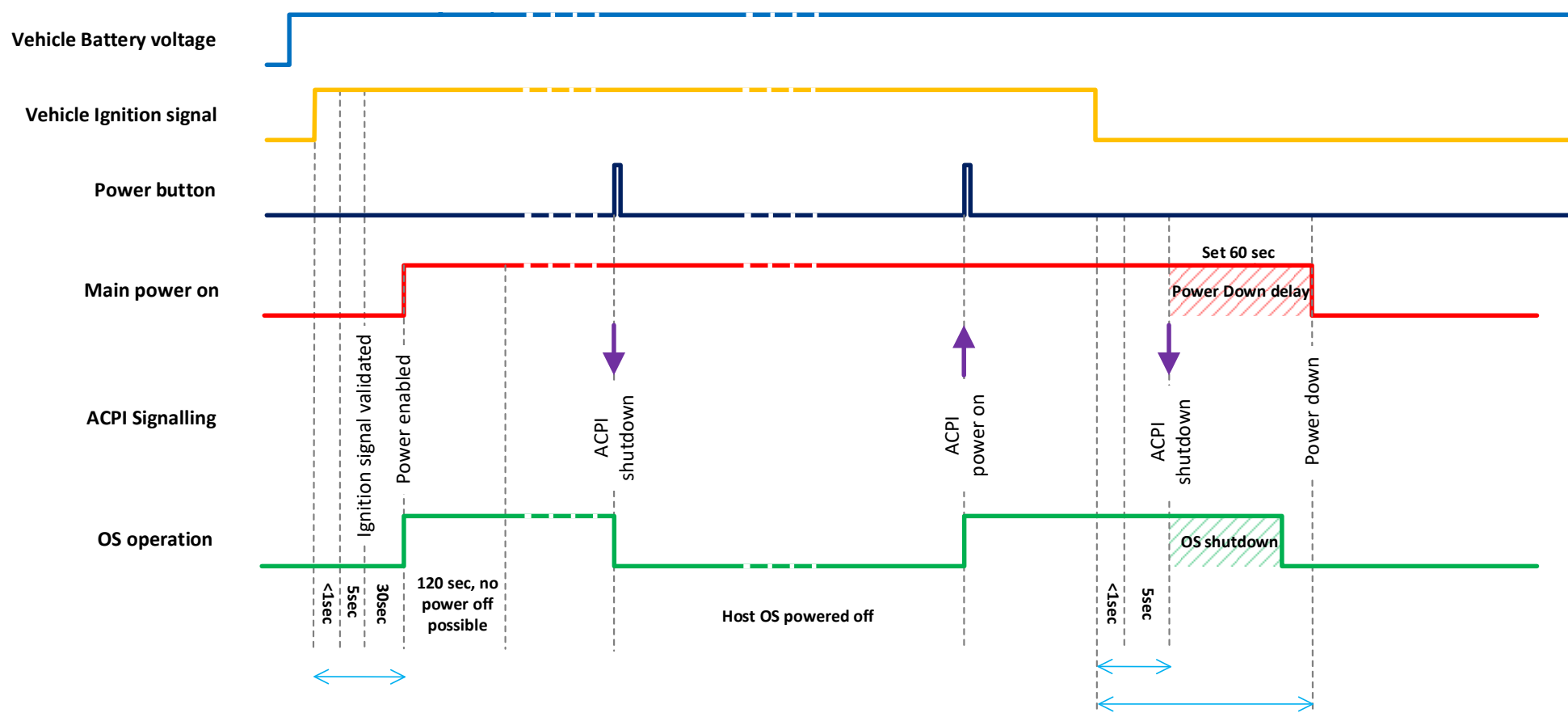
If the Ignition signal drops out for $\geq 5\text{sec}$ during operation of the CCU System, the system will power down with the configured power off delay and then restart.



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If the power button on the front of the CCU is pushed during operation of the CCU System, it will carry out an immediate graceful software shutdown. Power remains on the system. When the power button is pushed again the system will instantly boot again. The power button can be used during service of the system.



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